PHYS 357: Honours Quantum Physics 1

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Abstract

NOT FINISHED

Contents

1 Introduction

Textbook used is JS Townsend A Modern Approach to Quantum Mechanics, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on wednesdays, best 8 of 12).
- 20% midterms (best of 2, october 2nd and november 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

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2.1 Angular Momentum

Definition 1 (Orbital Angular Momentum)**.** *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \tag{1}$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x-y plane, only the z component survives. On a circular path of radius r with speed v,

$$L_z = mvr. \tag{2}$$

Definition 2 (Angular Momentum of a Rigid Body)**.** *For a rigid body rotating about a fixed axis this*

$$L = I_0 \omega, \tag{3}$$

with I_0 the moment of inertia and ω the angular velocity.

2.2 Magnetic moment of a circulating charge

Definition 3 (Magnetic moment)**.** *A magnetic moment is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

Consider a point particle of mass m and charge q in a circular orbit of radius r and period T. Three steps connect this to a magnetic moment.

1. The charge passes any point on the orbit once per period, so it is equivalent to a steady current

$$I = \frac{q}{T} = \frac{qv}{2\pi r}.$$

2. A current loop of area $A = \pi r^2$ has magnetic moment

$$\mu = \frac{IA}{c}.$$[1]

3. Substituting gives

$$\mu = \frac{1}{c}\left(\frac{qv}{2\pi r}\right)\left(\pi r^2\right) = \frac{q}{2mc}\, mvr = \frac{q}{2mc}\, L_z. \tag{4}$$

Result 1 (Gyromagnetic Ratio)**.** *The magnetic moment is proportional to the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}, \qquad \gamma = \frac{q}{2mc}. \tag{5}$$

The constant γ is the gyromagnetic ratio.

Remark 1. *This is a purely classical result. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

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2.3 Spin and the Magnetic Moment

Remark 2. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

In quantum mechanics the intrinsic angular momentum $\vec{S}$, called spin, replaces the orbital angular momentum $\vec{L}$. The magnetic moment is still proportional to it, but with an extra dimensionless factor,

$$\vec{\mu} = \frac{qg}{2mc}\, \vec{S}, \tag{6}$$

where g is the g-factor of the particle. It has no classical analogue and must be measured or derived from relativistic quantum theory.

Definition 4 (Gyromagnetic ratio)**.** *The gyromagnetic ratio is the ratio of the magnetic moment to the angular momentum,*

$$\gamma = \frac{qg}{2mc}, \tag{7}$$

where q is the charge, g is the g-factor, m is the mass, and c is the speed of light.

Definition 5 (Reduced Planck constant)**.**

$$\hbar = \frac{h}{2\pi}. \tag{8}$$

It carries units of angular momentum, so $[\hbar] = [L_z]$.

[1] Here c is in Gaussian units.

2.3.1 Magnetons

Factoring $\hbar$ out of the gyromagnetic ratio gives a natural unit of magnetic moment for each mass scale.

Definition 6 (Bohr magneton)**.**

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \tag{9}$$

Definition 7 (Nuclear magneton)**.**

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \tag{10}$$

The electron moment is then written as

$$\vec{\mu}_e = g_e\,\mu_B\,\frac{\vec{S}}{\hbar}, \tag{11}$$

and similarly for the proton and neutron with μ_N in place of μ_B.

2.3.2 Values for the three particles

Particle	Charge q	Mass m	g-factor	Gyromagnetic ratio γ
Electron	$-\lvert e\rvert$	m_e	$g_e \approx 2$	$g_e\mu_B/\hbar$
Proton	$+\lvert e\rvert$	m_p	$g_p \approx 5.59$	$g_p\mu_N/\hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n\mu_N/\hbar$

Table 1: Charges, masses, g-factors and gyromagnetic ratios.

Remark 3. *The neutron is neutral but still has a magnetic moment, so $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.*

Remark 4. *Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.*

2.4 Force in an Inhomogeneous Field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu}\cdot\vec{B}, \tag{12}$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U, \qquad F_z = -\frac{\partial U}{\partial z}. \tag{13}$$

Remark 5. *A uniform field gives no force, only a torque. The Stern-Gerlach apparatus needs a field* gradient *to deflect anything.*

2.4.1 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin.

Taking the field along z,

$$U = -\mu_z B_z = g_e \frac{|e|}{2m_e c} S_z B_z, \tag{14}$$

and therefore

$$F_z = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z, \tag{15}$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 2.

$$F_z \propto S_z. \tag{16}$$

The deflection of an atom measures the z component of its spin directly. Classically S_z varies continuously over $[-|\vec{S}|, |\vec{S}|]$, so the beam should smear into a band. It does not.

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$U = -\vec{\mu} \cdot \vec{B}$, $P_z = -\frac{dU}{dz} = \mu_z \frac{dB_z}{dz}$, $\mu_z = -\frac{-g_e|e|}{2m_e c} S_z$. Spin is a vector: $\vec{S} = (S_x, S_y, S_z)^T$. $S_z = |\vec{S}| cos\theta$, where $\theta \in [0, \pi] \implies cos\theta \in [-1, 1]$. Classically, S_z is continuous. Quantum mechanically, S_z is discrete. e-spin angular momentum is quantized: $S_z = \pm\hbar/2$. New notation a "ket" $|\uparrow\rangle$ and $|\downarrow\rangle$ for $S_z = +\hbar/2$ and $S_z = -\hbar/2$ respectively. The Stern-Gerlach experiment shows that the spin of an electron is quantized. The spin of an electron is a two-level system, so it can be represented as a two-dimensional vector space. The spin state of an electron can be represented as a linear combination of the basis states $|\uparrow\rangle$ and $|\downarrow\rangle$:

A Appendix

B Useful Links

Stern-Gerlach experiment: https://www.youtube.com/shorts/QG7kJXxN1Kk